Lab 2

Quality Control and Genome Assembly

Day 1 · 13:45–15:30 · 105 min
Pairs · half the room on the Illumina track, half on the Nanopore track

LO3: perform QC and assembly on raw reads in Galaxy, and judge whether an assembly is good enough to use

Open your handout

Lab 2 handout

`Lab2_qc_assembly.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

One isolate, sequenced twice

- An **MRSA** strain (KUN1163), sequenced on **Illumina** and on **Oxford Nanopore**
- Half the pairs take each track; at the end you compare

STRATEGY

Assembly takes **20–60 min**. Start it early and interpret QC while it runs.
Not finished by **15:00**? Use the fallback (Part E).

Plan for this lab

Part	What	Time
A	Set up and upload	10 min
B–D	Check quality → clean → start assembly	35 min
F	Interpret QC while it runs	20 min
G	Assess the assembly	15 min
H	Compare tracks	10 min

Upload your track's reads

1. New history: `Lab2 – Illumina` **or** `Lab2 – Nanopore`
2. **Upload** → **Paste/Fetch data** → paste the URL(s) **from your handout**
3. **Type:** `fastqsanger.bz2` → **Start**
4. Rename: `illumina_R1` + `illumina_R2`, **or** `nanopore_reads`

> **CHECK YOURSELF**
>
> Why are there two Illumina files but only one Nanopore file?

Illumina: check, clean, assemble

FastQC → **fastp** → **Shovill**

1. **FastQC** on `illumina_R1`, then on `illumina_R2`
2. **fastp:** Paired · R1 and R2 · defaults
3. **Shovill:** Paired-end · the two fastp outputs · defaults

> Shovill can’t see the fastp files? Pencil icon → **Datatypes** → `fastqsanger.gz`

Nanopore: check, filter, assemble

NanoPlot → **filtlong** → **Flye**

1. **NanoPlot** on `nanopore_reads`
2. **filtlong:** Min. length `1000` · defaults
3. **Flye:** filtlong output · Mode **Nanopore corrected** · defaults

Assembly running?

Don’t wait for it: go on to Part F and interpret your QC

20 min

Interpret your QC

	Your track
Number of reads	
Typical read length	
Total bases (≈ reads × length)	
Coverage = total bases ÷ 2.8 Mb	
Quality good enough? Why?	

Check yourself: quality

CHECK YOURSELF

1. Is your coverage above the minimum?
   (≈ 30× Illumina, ≈ 40–50× Nanopore)
2. Look at the report from a **bad** run on the screen. What looks different?

Assess the assembly with QUAST

Quast → your Shovill `contigs` or Flye `assembly` → defaults → HTML report

Metric	Good sign for *S. aureus*
# contigs (pieces)	Illumina: tens to ≈ 100 · Nanopore: 1–5
Total length	≈ 2.7–3.0 Mb
N50	Bigger is better
GC (%)	≈ 33%

Check yourself: assembly

CHECK YOURSELF

1. Is the total length close to the expected genome size?
2. What would a total length of **5.6 Mb** suggest?
3. Would you accept this assembly for AMR analysis? Why?

Compare the two tracks

	Illumina	Nanopore
# contigs		
Largest contig		
Total length		
N50		

- Which is more **complete**?
- Which is more **accurate**?
- Gene on a **plasmid**? Which would you prefer?
- What would a **hybrid** give you?

PART E · FALLBACK

Still running at 15:00?

1. Open the shared history link (handout Part E):
   usegalaxy.eu/u/aryakaul/h/ughe-ssi-course-precomputed
2. Click **Import**
3. Use the Shovill or Flye assembly and its QUAST report
4. Continue from Part G

> Your own job **keeps running**. Check it tonight!

Summary

Illumina → many accurate contigs

FastQC → fastp → Shovill → QUAST

Nanopore → few long contigs (≈ complete genome)

NanoPlot → filtlong → Flye → QUAST

Tomorrow: **what is in this genome?**

Adapted from the Galaxy Training Network “Bacterial Genome Assembly” tutorials (Shovill, Flye) (CC-BY 4.0).